

Nilay Singh

+91 7982631261 Patiala, India
nilay31singh@gmail.com LinkedIn GitHub LeetCode Portfolio

TECHNICAL SKILLS

PyTorch, Transformers, Librosa, FastAPI, React, SQLAlchemy, LangChain, PostgreSQL, Docker, Python

EXPERIENCE

- AI Engineering Intern** | [Live] [GitHub] June 2026 – July 2026
BEL Central Research Laboratory Ghaziabad, India
- Implemented TADN, a **~1.2M parameter Transformer model** for Multi-Object Radar Tracking (MOT) in high-density defense environments.
 - Achieved **92.12% data association accuracy** on holdout datasets, outperforming classical GNN baselines by **reducing track fragmentation from 527 to 0**.
 - Designed a sequence-to-sequence **Self/Cross-Attention architecture** to dynamically map track states and resolve overlapping aircraft geometries.
 - Devised a curriculum training strategy to ensure model robustness against severe radar latency (**up to 3.5s dropouts**) and extreme positional noise ($\sigma=140m$).
 - Evaluated an existing enterprise RAG chatbot utilizing the **RAG Triad framework**, subsequently introducing **Chain-of-Thought (CoT) prompting** to optimize contextual reasoning and reduce hallucination rates.

RESEARCH & PUBLICATIONS

- Green AI Cascade: Efficient Long-Form Video Inference** [Paper]
Accepted for Publication, CVIP 2026 (13.7% Acceptance Rate)
- Architected a **CPU-first pre-filtering pipeline** for long-form video analytics, reducing expensive LVL (Large Vision-Language Model) GPU workloads by **~30–45%**.
 - Conducted a **2,400-run grid search** evaluating spectral heuristics, discovering that standard face detection algorithms actively suppress target recall by **18.6 percentage points**.
 - Introduced YouTube 'Most Replayed' retention heatmaps as an **annotation-free behavioral ground truth**, eliminating the bottleneck of expensive manual data labeling.
 - Engineered a custom Temporal Non-Maximum Suppression (NMS) configuration to resolve over-merging, successfully preserving **up to 71.1% recall** of viral comedic moments.

PROJECTS

- SoulTune — AI Audio Analysis Pipeline** | *Python, FastAPI, Librosa, XGBoost, Gemini API* | [Live] [GitHub]
- Extracted **518 dynamic audio DSP features** (MFCCs, spectral centroids) via **Librosa** to establish a high-dimensional mathematical baseline for acoustic analysis.
 - Trained an **XGBoost classifier** achieving **61% accuracy** across **8 distinct genres**, integrating the **Gemini LLM API** to map audio parameters to generative visual profiles.
- Local RAG Chatbot — Privacy-First LLM Engine** | *FastAPI, FAISS, SentenceTransformers, Ollama* | [GitHub]
- Developed a local **RAG system** utilizing custom text chunking (**300-character windows, 50-char overlap**) to securely query raw document text with zero external API calls.
 - Engineered a retrieval pipeline mapping **384/768-dimensional embeddings** into a **FAISS** index, fetching the top-3 semantic matches to ground a local **Phi-3** model via **Ollama**.
- KHABRI — Autonomous NLP Agent** | *FastAPI, LangGraph, Tavily API* | [Live] [GitHub]
- Architected a **LangGraph state-machine agent** orchestrating a deterministic 3-node processing pipeline (search, summarize, output) to extract structured insights from news sources.
 - Integrated the **Tavily Search API** to fetch the top 10 relevant web results, processing them via an LCEL pipeline powered by **Gemini 2.5 Flash-Lite** ($T = 0.5$) within a **FastAPI** backend.

CERTIFICATIONS & ACHIEVEMENTS

- GATE DA 2025** — All India Rank **2804** in Data Science & Artificial Intelligence. 2025
- Machine Learning Specialization** — DeepLearning.AI & Stanford University [Credential] Jul 2025

EDUCATION

- Thapar Institute of Engineering and Technology** Patiala, India
Bachelor of Engineering, Computer Engineering | CGPA: 8.91/10 Aug 2023 – May 2027